# Robot Learning for Pick-and-Place: Imitation Learning with SmolVLA and ACT on a SO-101 Arm in Isaac Sim

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**Abstract**

We investigate imitation learning for a robotic pick-and-place task in which a SO-101 arm picks oranges and places them on a plate. Training data are collected inside NVIDIA Isaac Sim, edited manually with a custom GUI, and models are trained on the EPFL SCITAS cluster using the LeRobot framework. We compare two policy architectures—ACT (Action Chunking with Transformers) and SmolVLA (a compact Vision-Language-Action model)—and evaluate their success rates under various dataset sizes and training regimes. Our results show that ...

**Keywords:** robot learning, imitation learning, pick-and-place, Isaac Sim, ACT, SmolVLA, LeRobot.

# 1 Introduction

Manipulation tasks such as grasping and placing objects remain challenging for autonomous robots because they require precise perception, planning, and control under real-world variability. Imitation learning has emerged as an effective paradigm: a policy is trained from human (or simulated) demonstrations without hand-crafted reward engineering [].

In this project we study the pick-and-place of oranges using a **SO-101** 6-DOF robot arm. Demonstrations are generated inside **NVIDIA Isaac Sim**, edited with a custom annotation tool, and policies are trained with the **LeRobot** framework. We evaluate two architectures:

- **ACT** [] – Action Chunking with Transformers, a purely visuomotor policy.
- **SmolVLA** [] – a compact Vision-Language-Action model that conditions on a natural-language task description.

The remainder of this report is structured as follows. Section 2 reviews related work. Section 3 describes the experimental setup. Section 4 details training procedures. Section 5 presents quantitative and qualitative results. Section 6 concludes.

# 2 Related Work

## 2.1 Imitation Learning for Manipulation

## 2.2 Vision-Language-Action Models

## 2.3 Simulation-to-Real Transfer

# 3 Methodology

## 3.1 Robot and Simulation Setup

We use the **SO-101** arm simulated in **NVIDIA Isaac Sim**. The task is defined as picking a randomly placed orange from a table and placing it on a plate.

## 3.2 Dataset Collection

Demonstrations are recorded with `dataset_recorder.py`. Each episode stores RGB observations from two cameras and joint-position actions at 30.

## 3.3 Dataset Editing

A custom GUI (`dataset-editor/`) allows manual episode inspection, trimming, and relabelling. Episodes with failures or annotation errors are removed before training.

## 3.4 Policy Architectures

### 3.4.1 ACT

### 3.4.2 SmolVLA

## 3.5 Training Infrastructure

Models are trained on the EPFL SCITAS cluster using SLURM. Training scripts are located in `cluster-training/`.

# 4 Experiments

## 4.1 Evaluation Protocol

Each policy is evaluated over 20 rollouts per condition. A trial is counted as a success if the orange is placed on the plate within the episode time limit. Results are reported as success rate (%).

## 4.2 Baselines

- Pretrained SmolVLA (zero-shot, no fine-tuning).
- ACT trained from scratch on the full dataset.

## 4.3 Ablations

- Dataset size: 20 / 50 / 100 demonstrations.
- Tailed vs. standard dataset splits.

# 5 Results

## 5.1 Quantitative Comparison

Table 1: Success rates (%) on the pick-and-place task (20 trials each).

| Model | Dataset | Success Rate (%) |
|---|---|---|
| SmolVLA (pretrained, zero-shot) | – | – |
| ACT | full | – |
| SmolVLA fine-tuned | full | – |
| SmolVLA fine-tuned (tailed) | tailed | – |

## 5.2 Qualitative Analysis

# 6 Conclusion

## 6.1 Summary

## 6.2 Limitations

## 6.3 Future Work

# References